Arduino Project: Boolean Algebra Trivia Game

Everton Smith
Evertonwsmith.github.io/Everton/projects

Link: **https://www.tinkercad.com/things/2bUNRqS9FN4-trivia?sharecode=mQm1TOrS0dfiZms-TswxxG1CJVHEV9KaQx-uh86eyvs**

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Project Overview & Objectives

Learning project to enhance abilities in Arduino hardware setup and coding.

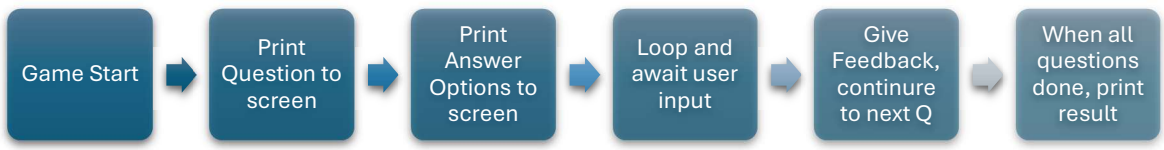
The goal was to build a multiple-choice trivia style game. Learning objectives included:

- Arduino/C++ code to await user input
- LCD display control via code
- Physical wiring of LCD display
- Real-time score tracking

System Architecture & Design

This system uses the Arduino Uno R3, coded in Arduino C++ language and hardwired via breadboard.

Figure 1: High-level logic plan



Push buttons will be used for user input, LCD 16x2 display for output, and RGB LED for visual feedback on answers. Arduino Uno R3 will be used, coded via Arduino IDE and USB interface in Arduino C++.

Components

Figure 2: Components

Name	Component	Quantity
U1	Arduino Uno R3	1
U2	LCD 16 x 2	1
R1	220 Ω Resistor	1
Rpot1	250 kΩ Potentiometer	1
S1, S2, S3	Pushbutton	3
D1	RCBG LED RGB	1
R2, R3, R4	0.22 kΩ Resistor	3

Schematic and Wiring Diagrams

Figure 3: Schematic diagram

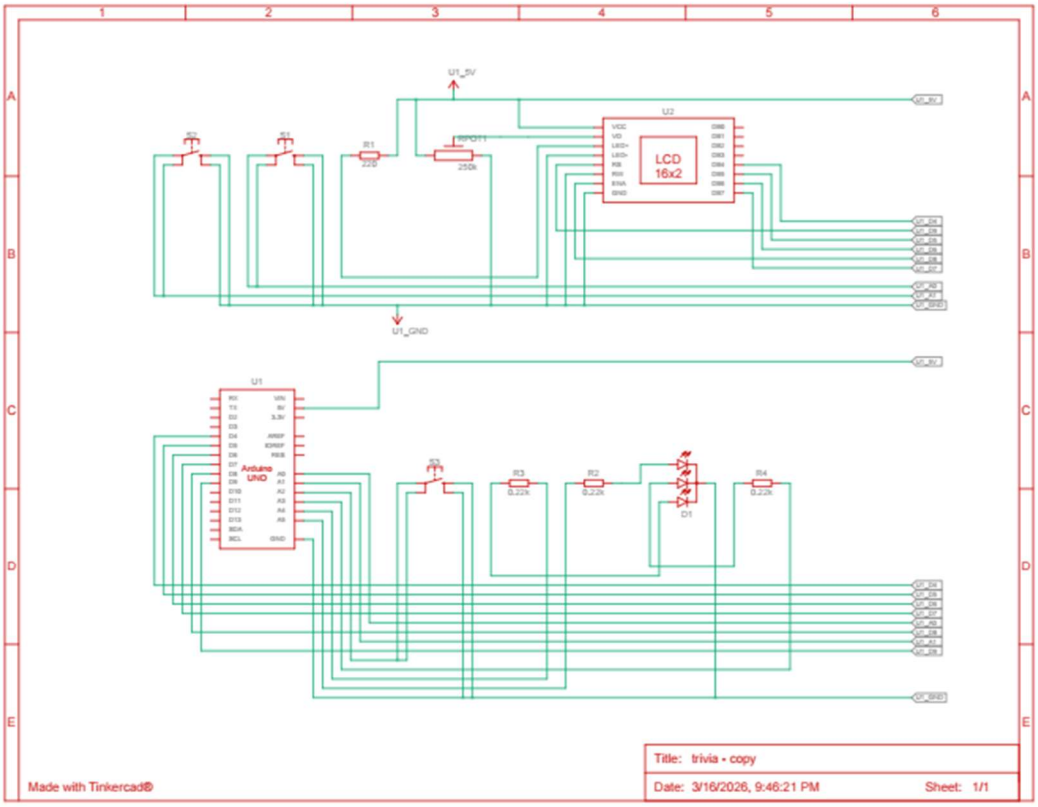
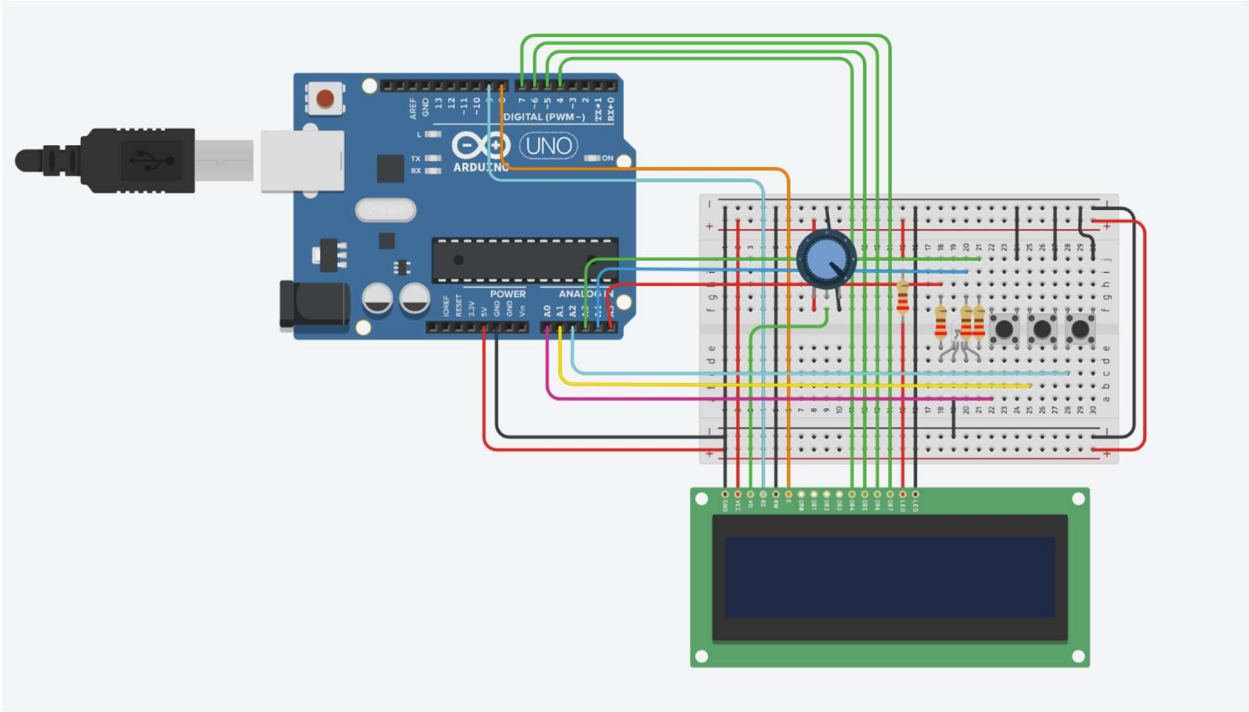


Figure 4: Wiring diagram



Software Logic

```
#include <LiquidCrystal.h>


//Some pin setup, constants, global variables, ELEMENTS function returns size of
//array

#define D4 4
#define D5 5
#define D6 6
#define D7 7
#define E 8
#define RS 9
#define ELEMENTS(x) (sizeof(x) / sizeof(x[0]))

const int ansA = A0;
const int ansB = A1;
const int ansC = A2;
const int red = A5;
```

```
const int blue = A4;
const int green = A3;
int qNum = 0;
int correct = 0;
int state = 1;
int finish = 0;
//Lists of questions and answers

char* questions[] = {
  "AB + A =",
  "A * 1 =",
  "AB + A'B =",
  "0 * 1 =",
  "C + C' =",
  "AA' ="
};
char* answers[] = {
  "A",
  "A",
  "B",
  "A",
  "B",
  "A"
};

char* answersText[] = {
  "A:A B:B C:1",
  "A:A B:B C:1",
  "A:A B:B C:1",
  "A:0 B:1 C:A",
  "A:0 B:1 C:C",
  "A:0 B:1 C:A"
};

LiquidCrystal LCD(RS, E, D4, D5, D6, D7);

void setup() {
  //initiate buttons for entering answers A, B, C respectively. (pins A0, A1, A2)
  pinMode(ansA, INPUT_PULLUP);
  pinMode(ansB, INPUT_PULLUP);
  pinMode(ansC, INPUT_PULLUP);

  //initiate rgb led outputs
  pinMode(red, OUTPUT);
  pinMode(blue, OUTPUT);
```

```
  pinMode(green, OUTPUT);

  //serial begin to allow waiting for input
  Serial.begin(9600);

  //initialize screen
  LCD.begin(16, 2);
}

void loop() {
  digitalWrite(green, LOW);
  digitalWrite(red, LOW);
  digitalWrite(blue, HIGH);
  LCD.print("Boolean Algebra Quiz");
  delay(1000);
  scroll(5);
  LCD.clear();
  digitalWrite(blue, LOW);
  startGame();

  LCD.clear();
}

void startGame() {
//State 1 is the introduction phase
  while (state == 1) {
    qNum = 0;
    correct = 0;
    state = 1;
    finish = 0;
    LCD.print("Choose A, B or C");
    delay(3000);
    LCD.clear();
    state = 0;
  }
//State 0 is main game state, loops through questions
  while (state == 0) {
    readQuestion();
  }
//State 2 is end of game and score summary
  while (state == 2) {
    LCD.print("You got ");
    LCD.print(correct);
    LCD.print(" /6");
    digitalWrite(green, HIGH);
```

```
    digitalWrite(red, LOW);
    delay(500);
    digitalWrite(green, LOW);
    digitalWrite(red, HIGH);
    delay(500);
    LCD.clear();
    finish++;
    if (finish == 6) {
      state = 1;
    }
  }
}

void readQuestion() {
  // 1. Display Question
  LCD.clear();
  LCD.print(questions[qNum]);
  delay(2000);

  // 2. Display Answers
  LCD.setCursor(0, 1);  // Move to second line
  LCD.print(answersText[qNum]);

  // 3. Wait for Input
  while (true) {
    // If using INPUT_PULLUP, check for LOW.
    // If using external resistors, stick with HIGH.
    bool btnA = (digitalRead(ansA) == LOW);
    bool btnB = (digitalRead(ansB) == LOW);
    bool btnC = (digitalRead(ansC) == LOW);

    if (btnA || btnB || btnC) {
      String currentCorrect = answers[qNum];
      bool isCorrect = false;

      if (btnA && currentCorrect == "A") isCorrect = true;
      if (btnB && currentCorrect == "B") isCorrect = true;
      if (btnC && currentCorrect == "C") isCorrect = true;

      LCD.clear();
      if (isCorrect) {
        digitalWrite(green, HIGH);
        LCD.print("Correct!");
        correct++;
      } else {
```

```
        digitalWrite(red, HIGH);
        LCD.print("Incorrect!");
      }

      delay(2000);
      digitalWrite(green, LOW);
      digitalWrite(red, LOW);

      // 4. NOW move to next question or end game
      qNum++;
      if (qNum >= ELEMENTS(questions)) {
        state = 2;  // Go to results screen
      }
      break;
    }
  }
}

void scroll(int num) {
  // scroll (num) to the left
  // to move it offscreen left:
  for (int positionCounter = 0; positionCounter < num; positionCounter++) {
    // scroll one position left:
    LCD.scrollDisplayLeft();
    // wait a bit:
    delay(500);
  }
  delay(1000);
}
```

Testing & Validation

Tested and validated via Tinkercad. Link at the top of this document

Figure 5: Question and Answer option readout

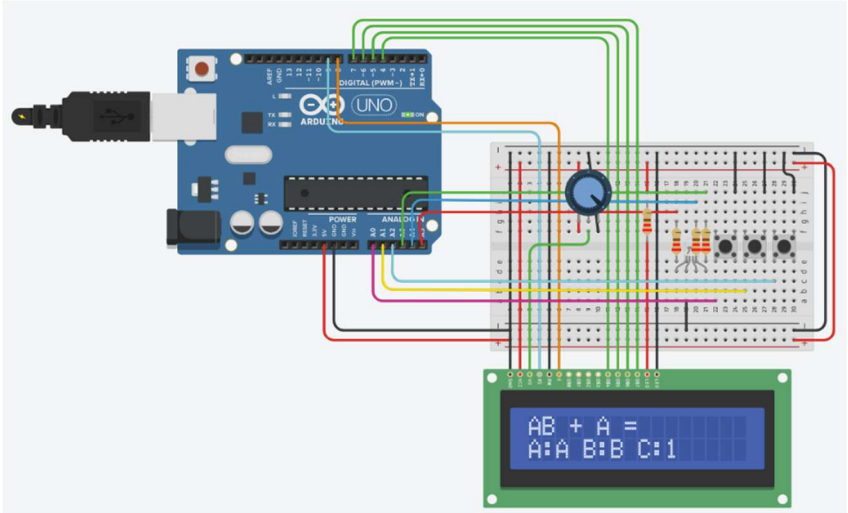
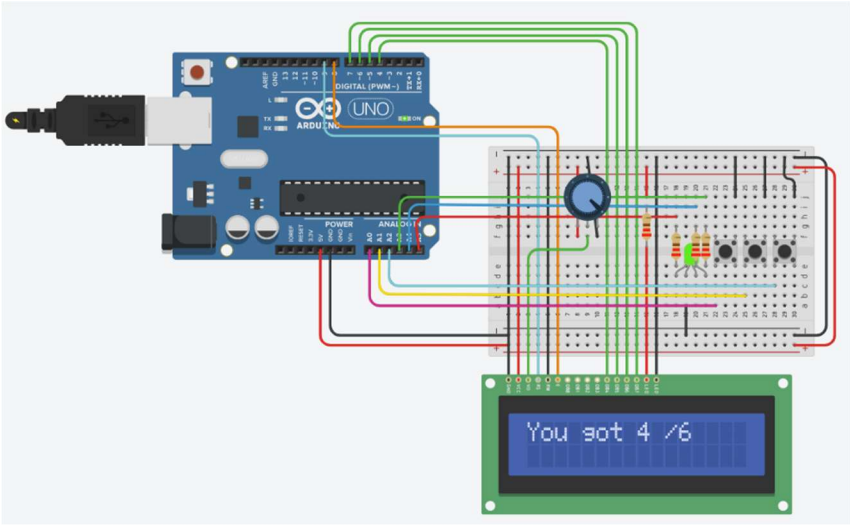


Figure 6: Results Readout



Conclusion & Future Improvements

This project in self learning was overall a success. I was able to implement the system I had imagined, using simple components and coding. Expanding on this I could do a number of things:

- Randomize questions
- Add more questions

- Randomize answer position
- Display ongoing score/progress

However, for the purposes of a simple demonstration, this project delivered all expectations.